

AI in Healthcare data

Data Preparation, Summarization, Visualization, Reduction and Diagnostic Performance Analysis of a Prostate Cancer Cohort

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AI in Healthcare Summer Camp – Data Preparation, Summarization, Visualization and Reduction

Abstract

Three raw datasets (demographic/clinical, lab values, comorbidity codes) were cleaned, merged and imputed into a single table (1,159 patients, 25 variables). PSA was evaluated as a metastasis marker via ROC (AUC=0.670). PCA reduced ten biomarkers to three components (69.3% variance). A logistic regression model (backward elimination, 10-fold CV) reached AUC 0.855 (accuracy 0.783), outperforming PSA alone. SVM and random forest performed similarly. LIME explained one patient's prediction. PSA is only a moderate discriminator; combining biomarkers, comorbidities and demographics improves prediction substantially.

Keywords: prostate cancer, PSA, ROC, PCA, logistic regression, cross-validation, LIME

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1. Introduction

Prostate cancer (PC) is common. Diagnosis relies on heterogeneous data: clinical records, lab tests, and medical history. Before any modelling, data must be cleaned, merged and understood. This report covers data preparation, visualization, feature reduction and classification. The main questions: (i) how to prepare three separate data sources into one reliable table, and (ii) how well does PSA alone discriminate metastasis, and can we compress the biomarker panel without losing too much information.

2. Data

Three source files, linked by patient ID:

- **DATASET1** – clinical: civil status, family cancer history, metastasis (outcome), dates (diagnosis, metastasis, birth).
- **DATASET2** – laboratory: ten numeric markers (monocyte, urea, lymphocyte, thrombocyte, erythrocyte, albumin, creatinine, globulin, PSA, coagulation factor).
- **DATASET3** – medical history: one row per condition (diabetes, hearing loss, hypercholesterolaemia, hypertension, osteoporosis, urinary retention).

3. Data Preparation and Results

3.1. Type correction and reshaping

Method: Converted categorical variables in DATASET1 to proper types; parsed dates. DATASET2 types were fine. DATASET3 was pivoted from long to wide format (one binary column per disease).

Result: All variables now in correct format for analysis.

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3.2. Missing data and imputation

Method: Checked missingness. DATASET1 had 63.7% missing in metastasis date (expected for non-metastatic patients). DATASET2 had scattered missingness (0.3–14.1%). Histograms (Fig. 1) showed most markers skewed, so we used median imputation.

Result: Missing data handled; no patient loss.

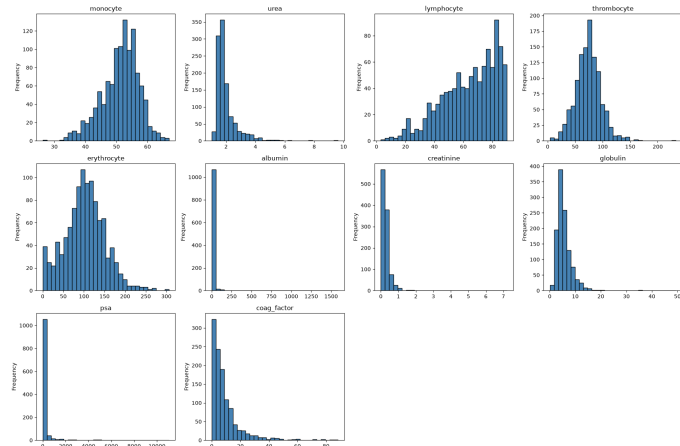


Figure 1: Distribution of ten biochemical markers.

3.3. Merging

Method: Left-joined DATASET1, DATASET2 and the wide disease table on ID. Filled missing disease flags with 0.

Result: Final merged dataset DATASET123: 1,159 rows $\times$ 25 columns.

3.4. Patient characteristics

Table 1 summarises the cohort.

Table 1: Patient characteristics (N = 1,159).

Variable	Value
Age at PC diagnosis (yr)	74.3 $\pm$ 8.7
Age at metastasis (yr)*	78.9 $\pm$ 8.5
Metastasis: yes	421 (36.3%)
Metastasis: no	738 (63.7%)
Family history: yes	623 (53.8%)
Civil status: married/cohabiting	808 (69.7%)

*only for patients with metastasis.

About one third developed metastasis, on average 4.6 years after initial diagnosis.

3.5. Correlation structure

Method: Computed Pearson correlations among the ten biomarkers and visualised as heatmap (Fig. 2).

Result: Most correlations weak; no strong collinearity. Fine to use all markers.

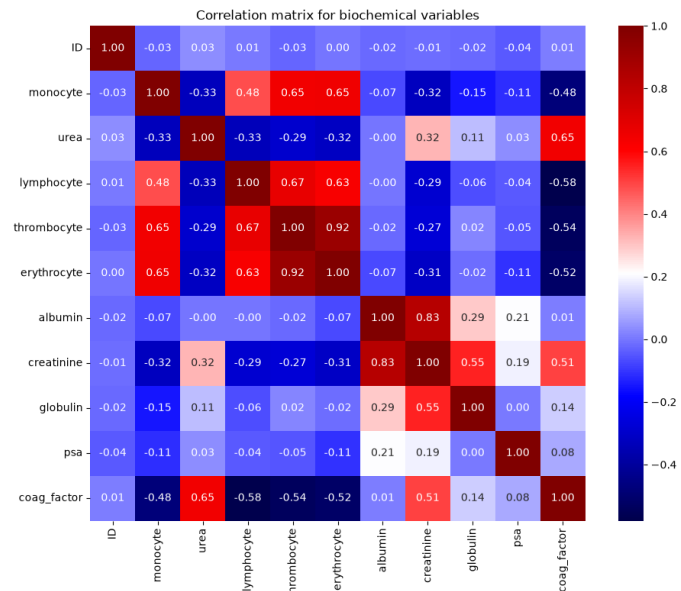


Figure 2: Correlation matrix of biochemical markers.

3.6. PSA diagnostic performance

Method: Tested two fixed cutoffs (20 and 100 ng/mL) for metastasis classification; also performed ROC analysis and found optimal cutoffs via Youden index and maximum accuracy.

Result: Table 2 shows performance. Lower cutoff gives higher sensitivity; higher cutoff gives higher specificity. Accuracy similar (0.65) due to class imbalance.

Table 2: PSA performance at fixed cutoffs.

	PSA >20	PSA >100
Sensitivity	0.596	0.323
Specificity	0.672	0.867
Accuracy	0.645	0.670

ROC curve (Fig. 3) gave AUC = 0.670 – moderate discrimination. Youden index chose 23 ng/mL; max accuracy chose 134 ng/mL.

ROC curve - PSA as predictor of metastasis

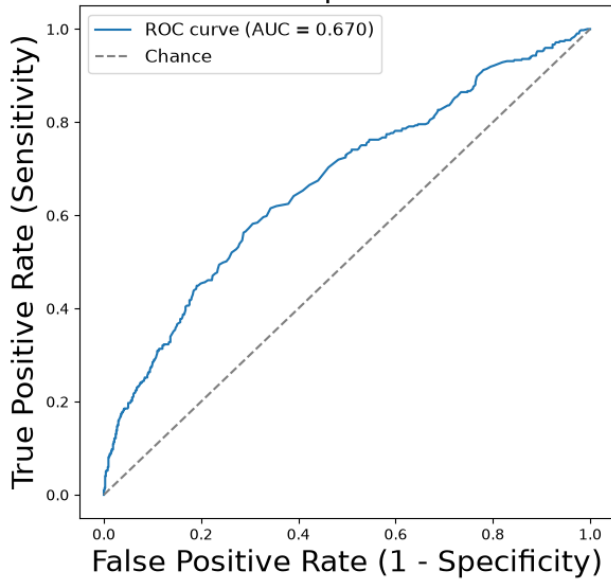


Figure 3: ROC curve for PSA (AUC=0.670).

3.7. Principal Component Analysis (PCA)

Method: Standardised biomarkers and performed PCA. Retained components with eigenvalue > 1 (Kaiser criterion).

Result: Three components retained (eigenvalues 3.90, 1.95, 1.10), explaining 69.3% of variance (Fig. 4). Loadings in Table 3. PC1: cellular markers, PC2: renal/nutritional, PC3: PSA-dominated. Good dimensionality reduction.

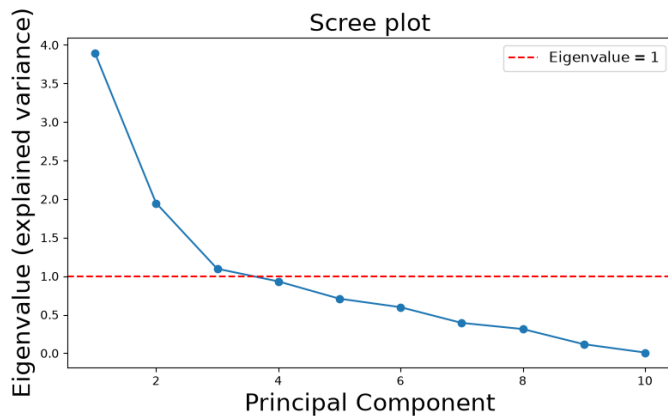


Figure 4: Scree plot with Kaiser cutoff.

Table 3: PCA loadings (top per component).

Marker	PC1	PC2	PC3
monocyte	0.381	0.112	-0.137
urea	-0.290	-0.041	-0.479
lymphocyte	0.348	0.167	0.051
thrombocyte	0.412	0.245	-0.225
erythrocyte	0.412	0.199	-0.260
albumin	-0.132	0.593	0.248
creatinine	-0.323	0.532	-0.047
globulin	-0.128	0.436	-0.345
psa	-0.085	0.173	0.600
coag_factor	-0.405	-0.063	-0.296

3.8. Univariate logistic regression (urinary retention)

Method: Fitted simple logistic regression: metastasis ~ urinary_retention.

Result: Significant (OR=2.49, $p < 0.001$) but low $R^2 = 0.029$. Accuracy at 0.5 cutoff: 65%, recall 37%.

3.9. Multiple logistic regression and collinearity

Method: Full model with all 21 predictors (excluding ID, dates, PCA components). Checked VIF.

Result: Pseudo $R^2 = 0.320$. High VIF for creatinine, albumin, coagulation factor (Table 4) – collinearity issue. Replacing biomarkers with PC1-3 solved VIF.

Table 4: VIF values (raw biomarkers).

Variable	VIF
Creatinine	63.4
Albumin	35.4
Coagulation factor	14.7
Globulin	6.6
Thrombocyte	5.3

3.10. Backward elimination

Method: Removed predictors one by one based on p-value until all remaining significant ($\alpha = 0.05$).

Result: Dropped 8 variables (including creatinine and coagulation factor). Final model had 13 predictors, AIC improved. In-sample performance at 0.5 cutoff: accuracy 0.79, recall 0.68, precision 0.72 (Table 5).

Table 5: Final model in-sample performance.

Class	Precision	Recall	F1
No metastasis	0.82	0.85	0.84
Metastasis	0.72	0.68	0.70
Accuracy	0.79		

3.11. Cross-validated performance

Method: 10-fold stratified cross-validation on final model.

Result: Mean metrics: accuracy 0.783 (SD 0.028), sensitivity 0.675 (SD 0.058), specificity 0.846, precision 0.718, F1 0.693, AUC 0.855 (SD 0.023). Stable. Huge improvement over PSA alone (AUC +0.185).

3.12. Model comparison

Method: Compared logistic regression, SVM, random forest, L1-logistic via same CV.

Result: All performed similarly (Table 6). Logistic regression has interpretable coefficients, so preferred.

Table 6: Cross-validated performance by model.

Model	Acc.	Sens.	Prec.	F1	AUC
Logistic Reg.	0.783	0.675	0.718	0.693	0.855
SVM	0.786	0.713	0.705	0.707	0.841
Random Forest	0.785	0.682	0.716	0.696	0.843
Reg. LR (L1)	0.783	0.677	0.717	0.694	0.855

3.13. LIME explanation

Method: Used LIME on one random patient to explain prediction.

Result: Top influential features: family history, albumin, diabetes, age, lymphocytes. LIME shows direction of each contribution for that patient.

4. Conclusion

Interpretation: We produced a clean merged dataset. PSA alone is a moderate discriminator (AUC 0.670), no single cut-off is perfect. PCA reduces ten markers to three components retaining 69% variance. A multivariable logistic model (13 predictors, cross-validated AUC 0.855) beats PSA significantly and remains interpretable. SVM and random forest give similar performance but are less transparent.

Impact: The logistic model can support risk stratification. The ROC analysis provides data-driven cutoffs when only PSA is available.

Limitations: Retrospective, cross-sectional design limits causal claims. Some issues: median imputation should be verified; PSA>20 flag duplicates continuous PSA; diabetes coefficient was negative (unexpected) – needs expert review. Future prospective studies with time-to-event data could use survival analysis.

5. Annexe

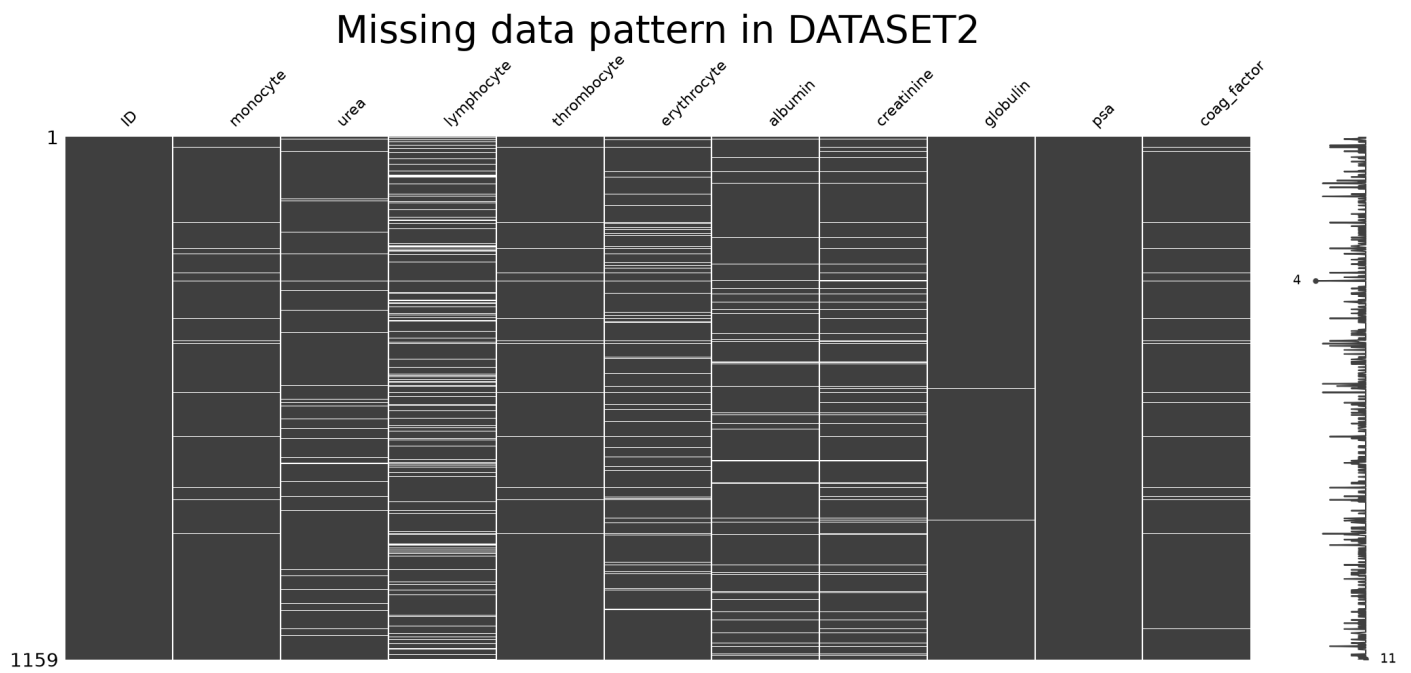


Figure 5: Missingness pattern in DATASET2 (scattered).